

7.1.5 Software Construction Process

NOTE — The Software Construction Process in this International Standard is a lower-level process of the Software Implementation Process.

7.1.5.1 Purpose

The purpose of the Software Construction Process is to produce **executable software units** that properly reflect the software design, ensuring correctness, consistency, and readiness for integration.

7.1.5.2 Outcomes

As a result of successful implementation of the Software Construction Process:

- a) verification criteria are defined for all software units against their requirements;
 - b) software units defined by the design are produced;
 - c) consistency and traceability are established between software units, requirements, and design; and
 - d) verification of the software units against the requirements and the design is accomplished.
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7.1.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Construction Process.

7.1.5.3.1 Software Construction

For each software item (or configuration item, if identified), the software construction activity consists of the following tasks:

7.1.5.3.1.1

The implementer shall develop and document the following:

- a) each software unit and database;
- b) test procedures and data for testing each software unit and database.

7.1.5.3.1.2

The implementer shall test each software unit and database to ensure that it satisfies its requirements. The test results shall be documented.

NOTE — Unit test evidence (e.g., Unit Test Reports, logs) shall be stored in /docs/ audit/construction/ or an equivalent controlled repository to maintain traceability with 7.1.5.5 (Audit and Configuration Control).

7.1.5.3.1.3

The implementer shall update the user documentation as necessary to reflect the software units produced.

7.1.5.3.1.4

The implementer shall update the test requirements and the schedule for Software Integration to reflect the constructed units.

Integration Readiness Checklist should include:

- 100 % unit test pass rate;
- All Non-Conformance Reports (NCRs) related to construction closed;
- Verification results signed by QA;
- Updated configuration baselines.

7.1.5.3.1.5

The implementer shall evaluate software code and test results considering the criteria listed below. The results of the evaluations shall be documented:

- a) traceability to the requirements and design of the software item;
 - b) external consistency with the requirements and design of the software item;
 - c) internal consistency between unit requirements;
 - d) test coverage of units;
 - e) appropriateness of coding methods and standards used;
 - f) feasibility of software integration and testing;
 - g) feasibility of operation and maintenance.
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7.1.5.3.1.6

The implementer shall conduct **peer reviews, code inspections, and static analysis** to identify defects, security vulnerabilities, and maintainability issues early. Code quality shall conform to organizational standards (e.g., maximum cyclomatic complexity thresholds). Findings and corrective actions shall be recorded and tracked.

7.1.5.4 Deliverables

The following deliverables shall be produced during this process:

- Source code and database artifacts for all software units;
- Unit test procedures and data;
- Unit Test Reports and logs;
- Code review and static analysis reports;
- Updated user documentation (if applicable);
- Integration readiness checklist and schedule updates;
- Construction Audit Log (CAL), documenting peer review findings and QA

approvals per build.

7.1.5.5 Audit and Configuration Control

All construction-related evidence (e.g., source code, test results, review logs, CAL entries) shall be placed under configuration management. Audit evidence shall be retained for at least one release cycle or as required by applicable regulatory or compliance policy. Digital signatures or equivalent controls shall be applied to ensure authenticity and integrity of records.